

Project Proposal Report

Real-Time Customer Support Ticket CRM using Apache Kafka and Agile Methodology

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1. Project Overview

This project is a Customer Relationship Management (CRM) web application where customers can create support tickets for their issues, and support agents can manage and resolve them.

The system will use Apache Kafka to enable real-time data streaming, so whenever a ticket is created or updated, it is instantly reflected across the system.

2. Objective (Goal)

- To manage customer issues efficiently
- To provide real-time ticket updates
- To implement Apache Kafka for event streaming
- To follow Agile methodology using sprints and user stories

3. Key Features

- User Registration and Login
- Create Support Ticket
- View Ticket Status (Open / In Progress / Closed)
- Agent Dashboard to manage tickets
- Real-time notifications using Kafka

4. System Workflow

Customer creates a ticket → Kafka sends event → System processes event → Agent dashboard updates → Agent resolves the issue → Status updated for customer

5. Agile Methodology (4 Sprints Plan)

Sprint 1: Basic Setup & Authentication

- User registration and login
- Basic UI design
- Database setup

Sprint 2: Ticket Management System

- Create and view tickets
- Store ticket data in database
- Basic ticket status tracking

Sprint 3: Kafka Integration

- Implement Kafka producer (send ticket events)
- Implement Kafka consumer (receive events)
- Real-time updates in dashboard

Sprint 4: Final Features & Testing

- Notification system
- UI improvements
- Testing and bug fixing

6. Sample User Stories

- As a customer, I want to create a support ticket so that my issue can be resolved.
- As a customer, I want to view my ticket status so that I know the progress.

- As a support agent, I want to see all tickets so that I can manage them.
- As a support agent, I want to update ticket status so that customers are informed.
- As a system, I want to send real-time updates using Kafka so that data is always up to date.

7. Tools & Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js / Python / Java (based on choice)
- Database: MySQL / MongoDB
- Messaging System: Apache Kafka
- Project Management Tool: Jira
- Methodology: Agile

8. Expected Outcome

- A fully functional CRM web application
- Real-time ticket updates using Kafka
- Proper Agile implementation with sprints and user stories
- A working project demonstration with dashboard

9. Conclusion

This project demonstrates how real-time systems can improve customer support using Apache Kafka and Agile practices. It provides a practical understanding of CRM systems, event-driven architecture, and project management tools like Jira.